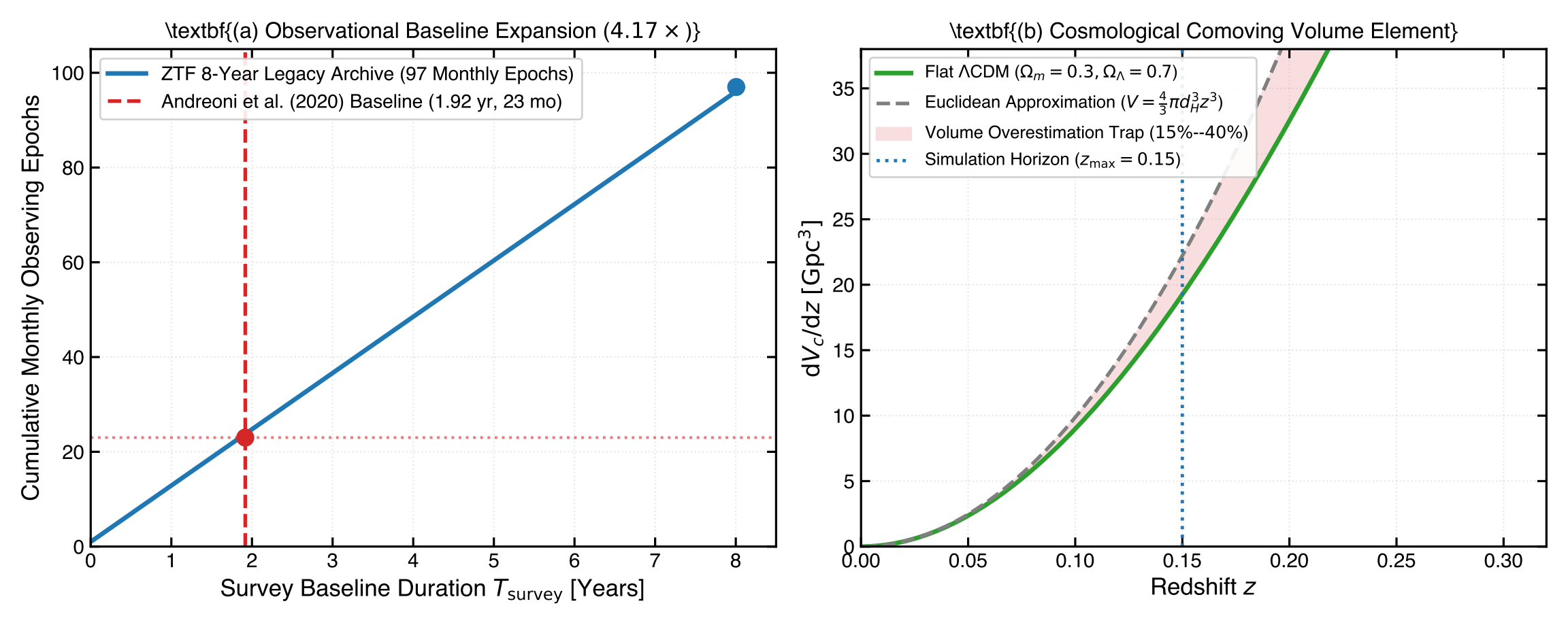
\textbf{(a) Observational Baseline Expansion (4.17 ×)}
ZTF 8-Year Legacy Archive (97 Monthly Epochs)
Andreoni et al. (2020) Baseline (1.92 yr, 23 mo)
Cumulative Monthly Observing Epochs
Survey Baseline Duration T_{survey} [Years]
0
20
40
60
80
100
0
1
2
3
4
5
6
7
8
\textbf{(b) Cosmological Comoving Volume Element}
Flat ΛCDM ($\Omega_m = 0.3, \Omega_\Lambda = 0.7$)
Euclidean Approximation ($V = \frac{4}{3}\pi d_H^3 z^3$)
Volume Overestimation Trap (15%--40%)
Simulation Horizon ($z_{\mathrm{max}} = 0.15$)
$\mathrm{d}V_c/\mathrm{d}z$ [Gpc^3]
Redshift z
0
5
10
15
20
25
30
35
0.00
0.05
0.10
0.15
0.20
0.25
0.30